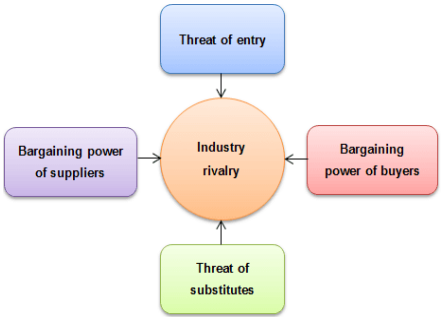
How do Porter's Competitive Forces Model help companies use information systems for competitive advantage?



New Market Entrants:

In a free economy with mobile labor and financial resources, new companies are always entering the marketplace. In some industries, there are very low barriers to entry, whereas in other industries, entry is very difficult. For instance, it is fairly easy to start a pizza business or just about any small retail business, but it is much more expensive and difficult to enter the computer chip business, which has very high capital costs and requires significant expertise and knowledge that is hard to obtain.

Substitute Products and Services:

In just about every industry, there are substitutes that your customers might use if your prices become too high. New technologies create new substitutes all the time. Internet telephone service has substituted for traditional telephone service, and streaming Internet music services for CDs, music stores, and digital download sites like iTunes. The more substitute products and services in your industry, the less you can control pricing and the lower your profit margins.

Customers:

A profitable company depends in large measure on its ability to attract and retain customers and charge high prices. The power of customers grows if they can easily switch to a competitor's products and services or if they can force a business and its competitors to compete on price alone in a transparent marketplace.

Suppliers:

The market power of suppliers can have a significant impact on firm profits, especially when the firm cannot raise prices as fast as suppliers can. The more suppliers a firm has, the greater control it can exercise over those suppliers in terms of price, quality, and delivery schedules.

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Strategies formulated using Information Systems:

Low-Cost Leadership:

By keeping prices low and shelves well stocked using a legendary inventory refill system, Walmart became the leading retail business. Point-of-sale terminals record the bar code of each item passing the checkout counter and send a purchase transaction directly to a central computer at Walmart headquarters. The computer collects the orders from all stores and transmits them to suppliers. Suppliers can also access store's sales and inventory data by using web technology.

Because the system refill inventory with lightning speed, Walmart does not need to spend much money on maintaining large inventories of goods in its own ware houses. The system also enables store to adjust the items stocked in its stores to meet customer demands. By using systems to keep operating costs low, store can charge less for its products than competitors yet reap higher profits.

Product Differentiation:

Apple has continued to differentiate its handheld computing products with nearly annual introductions of new iPhone and iPad models.

For example, Nike sells customized sneakers through its NIKE BY YOU program on its website. Customers can select the type of shoe, colors, material, outsoles, and laces. Nike transmits the orders by computers to specially equipped plants in China and Korea. The sneakers take only about three weeks to reach the customer.

Focus on Market Niche:

The data come from a range of sources. Credit card transactions, demographic data, purchase data from checkout counter scanners at supermarkets and retail stores, and data collected when people access and interact with websites.

Sophisticated software tools find patterns in these large pools of data and infer rules from them that can be used to guide decision making. Analysis of such data drives one-to-one marketing by which personal messages can be created based on individualized preferences.

For example, Hilton Hotel system analyzes detailed data collected on active guests in all of its properties to determine the preferences of each guest and each guest's profitability. Hotel uses this information to give its most profitable customers additional privileges, such as late checkouts. Contemporary customer relationship management (CRM) systems feature analytical capabilities for this type of intensive data analysis.

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Strengthen Customer and Supplier Intimacy:

Use information systems to tighten linkages with suppliers and develop intimacy with customers. Toyota, Ford, and other automobile manufacturers have information systems that give their suppliers direct access to their production schedules, enabling suppliers to decide how and when to ship supplies to the plants where cars are assembled. This allows suppliers more lead time in producing goods.

On the customer side, Amazon.com keeps track of user preferences for all the products it sells, including book and music purchases and can recommend titles purchased by others to its customers. Strong linkages to customers and suppliers increase switching costs.

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